

2.1: Notations and Visualizations

Longitudinal analysis:

- repeated observations taken on individuals.

Primary goals:

- ① characterize the change in responses or measurements over time.
- ② Identify factors that influence this change.

correlation: observations that belong to the same subject tend to be more similar than observations from different subjects. This must be accounted for in order to obtain valid inference.

Longitudinal data is a special form of hierarchical data, where observations are nested within different hierarchical/ordered levels.

Notation:

• y_{ij} = response variable for the i^{th} individual ($i = 1, \dots, N$) measured at time t_{ij} ($j = 1, \dots, n$)

• $\mu_j = E(y_{ij})$ = mean of y_{ij} & $\bar{y}_j$ = average of the response variable of all individuals @ time j .

- $\sigma_j^2 = \text{var}(Y_{ij}) \quad \forall \quad i = 1 \text{ to } N.$
- $\sigma_{jk} = \text{cov}(Y_{ij}, Y_{ik})$, covariance of Y_{ij} and Y_{ik}
- $r_{jk} = \frac{\sigma_{jk}}{\sigma_j \sigma_k} = \text{correlation btwn } Y_{ij} \text{ and } Y_{ik},$
ranges $[-1, 1]$. 0 = independent.
- $Y_i = (Y_{i1}, \dots, Y_{in})^T$ vector of repeated measures for person i ,
- $\text{COV}(Y_i) = \sum_i = \begin{bmatrix} \sigma_{11}^2 & \sigma_{12}^2 & \dots & \sigma_{1n}^2 \\ \sigma_{12}^2 & \sigma_{22}^2 & & \\ \vdots & & \ddots & \\ \sigma_{1n}^2 & & & \sigma_{nn}^2 \end{bmatrix}$ // symmetric matrix
for individual i ,
across all time points
from 1 to n .
- correlation matrix =
 $r_i = \begin{bmatrix} 1 & r_{12} & \dots & r_{1n} \\ r_{12} & 1 & & \\ \vdots & & \ddots & \\ r_{1n} & & & 1 \end{bmatrix}$ // for person i

For longitudinal data: it is often the case that:

- correlations are **positive**
- correlations decrease with increasing time separation
- **correlations btwn repeated measures rarely ~ 0 .**
- correlations btwn repeated measures taken very closely together in time rarely ~ 1

We will talk about 2 models:

- 1) LMM: used to analyze data that have both fixed and random effects. This is useful in analyzing hierarchical, longitudinal, or correlated datasets.
- 2) GEE: used to analyze correlated data, focusing on population-average effects.

2.2: Linear Mixed Model

Recall: for person i , we have $y_i = (y_{i1}, \dots, y_{in})^T$, which is a vector of repeated measures over time.

$$\text{cov}(y_i) = \Sigma_i = \begin{bmatrix} \sigma_1^2 & & & \\ \sigma_{12} & \sigma_2^2 & & \\ \vdots & & \ddots & \\ \sigma_{1n}^2 & & & \sigma_n^2 \end{bmatrix} \quad // \text{Symmetric matrix.}$$

3 assumptions:

• Assume model $E(y_i) = X_i \beta$, where X_i = vector of covariates

• Assume $y_i \sim$ multivariate normal,

with $\text{cov}(y_i) = \Sigma_i = \Sigma_i(\theta)$, where θ = vector of covariance.

• Assume some structure for $\Sigma_i(\theta)$, which can be:

- unstructured: does not assume anything about the terms of the covariance matrix Σ_i which is a total of $n(n+1)/2$ terms.

- compound symmetry: assume variance is constant across time:

$$\Sigma_i = \sigma^2 \begin{bmatrix} 1 & \rho & \dots & \rho \\ \rho & 1 & & \\ \vdots & & \ddots & \\ \rho & \dots & & 1 \end{bmatrix} \quad // \text{need to estimate } \sigma \text{ and } \rho$$

- Autoregressive: assume variance is constant across time and covariance is:

$$\Sigma_i = \sigma^2 \begin{bmatrix} 1 & \rho & \rho^2 & \dots & \rho^{n-1} \\ \rho & 1 & & & \rho^{n-2} \\ \vdots & & \ddots & & \\ \rho^{n-1} & \rho^{n-2} & \dots & & 1 \end{bmatrix} \quad // \text{still need to estimate } \sigma \text{ and } \rho.$$

with these we estimate β using MLE. The estimator, $\hat{\beta}$, is the generalized least square estimator!

$$\hat{\beta} = \left\{ \sum_{i=1}^N (X_i^T \Sigma_i^{-1} X_i) \right\}^{-1}$$

we use Restricted maximum likelihood (REML) bc the traditional MLE is biased downward,

Modeling the covariance (Σ_i) matrix: 3 ways:

- | | |
|---|--|
| 1) Assume unstructured; | } covariance pattern model.
Does not differentiate b/wn-
and within-subject variability, |
| 2) Assume a covariance structure | |
| 3) use random effects covariance structure, | |

Focus on approach 3, bc LMM uses it.

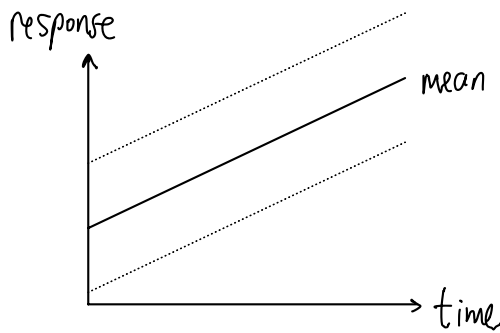
In LMM:

- Each individual have their own subject-specific mean response trajectory over time.
- Mean response = combination of:
 - fixed effects (shared by all individuals)
 - random effects (subject-specific, aka b/wn-subject effects),

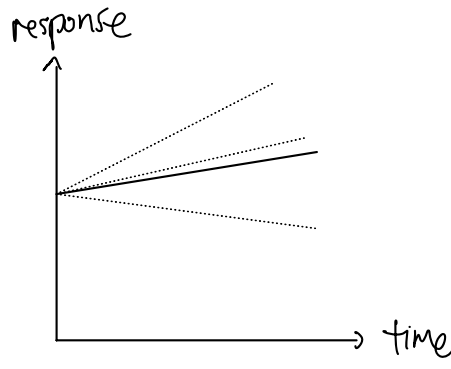
$\Rightarrow$ LMM = particular type of hierarchical model which contains both fixed and random effects.

Random Intercepts Model:

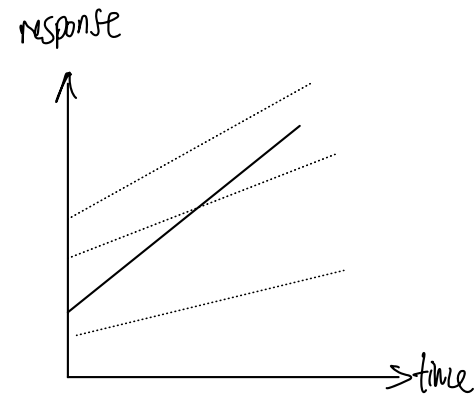
You let the y-intercept be different for different individuals.



random intercept



random slope



random intercept + slope

Consider the model:

$$Y_{ij} = \underbrace{X_{ij} \beta}_{\text{fixed effects}} + \underbrace{b_i}_{\text{random subject effect}} + \underbrace{\epsilon_{ij}}_{\text{residual error}}, \text{ where:}$$

- β is fixed effects
- b_i = random subject-specific (between-subject) effects,
 $b_i \sim N(0, \sigma_b^2)$
- ϵ_{ij} = residual error / within-subject error,
 $\epsilon_{ij} \sim N(0, \sigma^2)$
- $b_i \perp \epsilon_{ij}$

Interpretation:

- $X_{ij}\beta$ = fixed effects
 - response for the i^{th} person at the j^{th} time differs from population mean $X_{ij}\beta$ by 2 different effects:
 - subject effect b_i .
 - within-subject error ϵ_{ij} .
- } both are r.v.

want to estimate:

b_i , which is a r.v. $\sim N(0, \sigma_b^2)$, so we need to estimate σ_b^2 .

Conditional mean response: is the mean response trajectory over time for an individual i is:

$$E(Y_{ij} | b_i) = X_{ij}\beta + \underline{b_i} \quad // \text{has } b_i \text{ term}$$

$\uparrow$
individual i

Marginal mean response: is the mean response profile for the population:

$$E(Y_{ij}) = X_{ij}\beta$$

The random effect b_i reflects an individual's deviation from the population mean intercept, after the effects of the covariates have been accounted for.

This model induces the following compound symmetry pattern:

$$\Sigma_i = \begin{bmatrix} \sigma_{\mu_0}^2 + \sigma^2 & \sigma_{\mu_0}^2 & \dots & \sigma_{\mu_0}^2 \\ \sigma_{\mu_0}^2 & \sigma_{\mu_0}^2 + \sigma^2 & & \\ \vdots & & \ddots & \\ \sigma_{\mu_0}^2 & & & \sigma_{\mu_0}^2 + \sigma^2 \end{bmatrix}$$

proof:

$$\begin{aligned} \text{var}(Y_{ij}) &= \text{var}(X_{ij}\beta + b_i + \Sigma_{ij}) \\ &= \text{var}(b_i + \Sigma_{ij}) \quad // X_{ij}\beta \text{ is not r.v.} \\ &= \text{var}(b_i) + \text{var}(\Sigma_{ij}) \\ &= \sigma_b^2 + \sigma^2 \end{aligned}$$

$$\begin{aligned} \text{cov}(Y_{ij}, Y_{ik}) &= \text{cov}(X_{ij}\beta + b_i + \Sigma_{ij}, X_{ik}\beta + b_i + \Sigma_{ik}) \\ &= \text{cov}(b_i + \Sigma_{ij}, b_i + \Sigma_{ik}) \\ &= \text{cov}(b_i, b_i) + \cancel{\text{cov}(b_i, \Sigma_{ij})}^\perp + \cancel{\text{cov}(\Sigma_{ik}, b_i)}^\perp + \cancel{\text{cov}(\Sigma_{ij}, \Sigma_{ik})}^\perp \\ &= \sigma_b^2 \end{aligned}$$

σ_b^2 and σ^2 are "variance components". we use them to get β , which we do care about.

Intraclass Correlation (ICC): is the proportion of the total variance that is "explained" by between-subject variability.

$$ICC = \frac{\sigma_b^2}{\sigma_b^2 + \sigma^2}, \in [0, 1].$$

Notes on Estimation:

- we imposed normal distribution assumptions, so we can write out the likelihood function,
- degree of freedom is tricky. use "satterthwaite's" method.

Time Effect: using random intercepts model,

- How does the response change over time?

Is the mean response varying with time?

$$\begin{aligned} Y_{ij} &= X_{ij} \beta + b_i + \varepsilon_{ij} \\ &= \beta_0 + \beta_1 x_{ij1} + \beta_2 x_{ij2} + \beta_3 x_{ij3} + b_i + \varepsilon_{ij} \end{aligned}$$

For example, using dental dataset:

$$x_{ij1} = 1 \text{ for age 10}$$

$$x_{ij2} = 1 \text{ for age 12}$$

$$x_{ij3} = 1 \text{ for age 14}$$

$$H_0: \beta_1 = \beta_2 = \beta_3 = 0. \text{ (No time effect)};$$

Group Effects

• Is the change of the mean response different in groups?

For example, btwn boys and girls.

consider the model with a term for sex:

$$Y_{ij} = \beta_0 + \beta_1 X_{ij1} + \beta_2 X_{ij2} + \beta_3 X_{ij3} + \beta_4 X_{ij4} + b_i + \varepsilon_{ij}$$

where $X_{ij4} = 1$ for boys, reference (0) = girls.

Difference in mean response btwn girls and boys:

$$E(Y_{ij} | X_{ij4} = 1) - E(Y_{ij} | X_{ij4} = 0) = \beta_4$$

Interaction btwn time and group:

"How does the relationship of response wrt time (dental growth vs time) vary by group (gender)?"

$$Y_{ij} = \beta_0 + \beta_1 X_{ij1} + \beta_2 X_{ij2} + \beta_3 X_{ij3} + \beta_4 X_{ij4} + b_i + \varepsilon_{ij} \\ + \beta_5 X_{ij1} X_{ij4} + \beta_6 X_{ij2} X_{ij4} + \beta_7 X_{ij3} X_{ij4}$$

interaction terms

$$H_0: \beta_5 = \beta_6 = \beta_7 = 0.$$

Linear Time:

using a linear fit btwn the mean response and the time, instead of using indicators for time.

Let's assume a linear relationship btwn the time variable and the mean response:

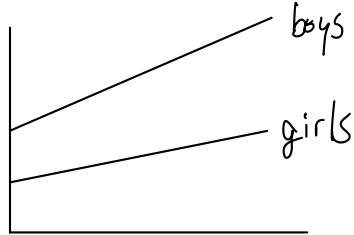
$$Y_{ij} = \beta_0 + \beta_1 \underbrace{X_{ij1}}_{\substack{\text{indicator} \\ \text{for sex}}} + \beta_2 \underbrace{X_{ij2}}_{\substack{\text{age, as} \\ \text{a number}}} + \beta_3 \underbrace{X_{ij1} X_{ij2}}_{\substack{\text{interaction} \\ \text{term}}} + b_i + \epsilon_{ij}$$

// intercept β_0 = mean distance among girls @ age 0.

// coefficient β_1 = the difference btwn boys vs girls @ age 0.

// coefficient of age = β_2 = change in mean response for every 1 year increment of age among girls (reference).

// coefficient of interaction term = β_3 = additional change of the regression coefficient for age in boys.



Predicting Random Effects

Subject-specific response profiles:

It is possible to obtain predictions of the subject-specific effects, b_i , and then of the subject-specific response trajectories $X_i\beta + b_i$.

$\hat{\beta}$ = estimate of β = fixed effects

$\hat{b}_i$ = conditional mean estimate of b_i (subject specific effects)

= conditional mean of b_i given the data Y_i

= aka BLUP (best linear unbiased predictor)

= "empirical Bayes" estimator since $\hat{b}_i$ can be derived from a fully Bayesian formulation.

= "shrunk" or "shrinkage effect".

$$\Rightarrow b_i = \frac{\sigma_b^2}{\sigma_b^2 + \sigma^2/n} \left(\underbrace{\bar{y}_i}_{\text{overall mean for person } y} - \underbrace{\bar{y}_{..}}_{\text{overall mean of all } y_s} \right)$$

Random Intercepts and Slopes Model

This model assumes that individuals vary in both the baseline level of response (intercept), and in their changes (slope) in the mean response over time:

$$Y_{ij} = B_0 + X_{ij} \beta + b_i + \alpha_i X_{ij} + \epsilon_{ij}$$

b_i = random intercept

α_i = random slope

ϵ_{ij} = residual error, $\sim N(0, \sigma^2)$

$\alpha_i, b_i \perp \epsilon_{ij}$

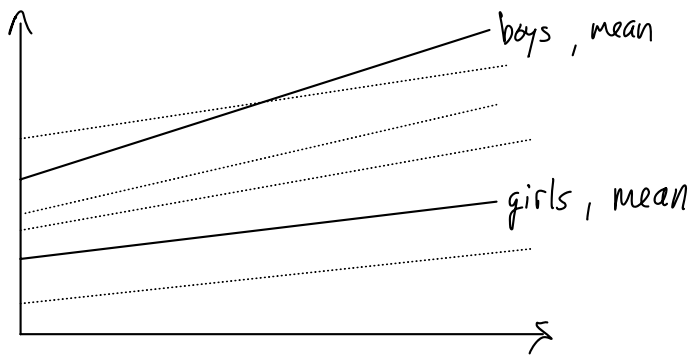
$$\begin{pmatrix} b_i \\ \alpha_i \end{pmatrix} \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_b^2 & \sigma_{b\alpha} \\ \sigma_{b\alpha} & \sigma_\alpha^2 \end{pmatrix} \right)$$

conditional mean response: for a specific individual

$$E(Y_{ij} | b_i, \alpha_i) = (B_0 + b_i) + (B + \alpha_i) X_{ij}$$

Marginal mean response: is averaged over all individuals in the population.

$$E(Y_{ij}) = \beta + 0 + X_{ij} \beta$$



2.3 : Generalized Estimation Equations (GEE)

For the linear regression models for continuous responses above, interpretation of β is independent of assumptions about correlation among the repeat measures.

This is not the case for discrete data.

$$E(Y_i) = X_i \beta$$

$$E(Y_i | b_i) = X_i \beta + b_i$$

$$\begin{aligned} E(Y_i) &= E(\underbrace{E(Y_i | b_i)}_{X_i \beta + b_i}) = E(X_i \beta + b_i) \\ &= X_i \beta + \underbrace{E(b_i)}_0 \\ &= X_i \beta \end{aligned}$$

Let's say you have a logistic or Poisson regression:

$$h(X_i \beta + b_i)$$

$$E\{h(X_i \beta + b_i)\} \neq h(X_i \beta + E(b_i))$$

Marginal Models: GEEs are known as Marginal Models.

- Marginal models = "population-average models".
- Indicates that the model for the mean response only depends on covariates, not random effects.
- Do not require distribution assumptions, only a regression model for the mean response.
- Avoidance of distributional assumptions leads to GEEs

Marginal model for longitudinal data has 3 parts.

① The conditional expectation (mean of each response):

$E(Y_{ij} | X_{ij}) = \mu_{ij}$, depends on the covariates thru a linked function:

$$g(\mu_{ij}) = \eta_{ij} = X_{ij} \beta$$

② The conditional variance:

$$\text{var}(Y_{ij}) = \phi v(u_{ij}), \text{ where:}$$

- $v(\mu_{ij})$ is a known "variance function",

- ϕ is a scale parameter that may be known or may need to be estimated.

③ conditional within-subject association: among repeated measures, given the covariates, is assumed to be a function of an additional set of association parameters, α .

$$\boxed{\text{corr}(Y_{ij}, Y_{ik}) = \alpha.}$$

There is no assumption of distribution. There will be no MLE.

Scenario : Response is 0/1 .

Bernoulli r.v is used,

$$E(Y_{ij}) = \mu_{ij} = P(Y_{ij} = 1)$$

$$(1) \text{ logit}\left(\frac{\mu_{ij}}{1 - \mu_{ij}}\right) = \eta_{ij} = X_{ij} \beta$$

$$(2) \text{ var}(Y_{ij}) = \underbrace{\mu_{ij}(1 - \mu_{ij})}_{\phi = 1 \quad V(\mu_{ij})}$$

$$(3) \log \text{ OR}(Y_{ij}, Y_{ik}) = \alpha$$

The GEE estimator of $\beta \Rightarrow$ minimizing the objective function wrt β :

$$\min_{\beta} \sum_{i=1}^N \left\{ Y_i - \underbrace{\mu_i(\beta)}_{=g^{-1}(X_{ij}\beta)} \right\}^T \underbrace{V_i^{-1}}_{\text{working covariance matrix}} \left\{ Y_i - \mu_i(\beta) \right\}, \quad \text{where}$$

• V_i = "working" covariance matrix

• μ_i = vector of mean responses with elements

$$\mu_{ij} = \mu_{ij}(\beta) = g^{-1}(X_{ij} \beta)$$

$$\Rightarrow \sum_{i=1}^N D_i^T V_i^{-1} (Y_i - \mu_i) = 0, \quad // \text{ Generalized Estimation Equations}$$

where $D_i = \frac{\partial \mu_i}{\partial \beta}$ is the derivative matrix .

Difference in interpretation: using an example,
response: $Y = 1$ or $Y = 0$ (abnormal vs normal)
predictor: $X = 1$ or $X = 0$ (drug vs placebo)
time points: 1, 2.
subjects: 67.

Consider the marginal model (using GEE to estimate):

$$\text{logit}(u_{ij}) = \beta_0 + \beta_1 X_{ij1} + \beta_2 X_{ij2}, \text{ where}$$

$X_{ij1} = 1$ for the active drug, 0 for placebo

$X_{ij2} = 0$ if time point 1, 1 if time point 2.

The within-subject association btwn the responses is modeled in terms of a common log odds ratio:

$$\alpha = \log \text{OR}(Y_{i1}, Y_{i2}).$$

The estimate for β is $\hat{\beta}_1 = 0.57$. That can be interpreted as: the odds of an abnormal outcome is $1.77(e^{0.57})$ times higher when treated compared to placebo.

Now, using generalized LMM:

$$\text{logit}\{E(Y_{ij} | b_i)\} = \beta_0^* + \beta_1^* X_{ij1} + \beta_2^* X_{ij2} + b_i,$$

where the random effects $b_i \sim N(0, \sigma_b^2)$

The estimate for β_1^* is $\beta_1^* = 0.89$ $\swarrow = 2.4$

$\Rightarrow$ odds of abnormal outcome is $e^{0.89}$ times higher when treated compared to placebo.

In this model, each patient is assumed to have a underlying propensity for an abnormal EKG given by b_i .

How to choose mixed model vs GEE?

- If you want to describe the between vs within variance, then use LMM.
- Subject specific trajectories: LMM.
- Avoid making distributional assumptions: GEE,
- Only care about effects of covariates on the population mean response: GEE,
- For non-linear model: Interpretation is different!

2.4 : Joint Modeling

use joint modeling when there is longitudinal data as well as time-to-event outcome.

There are 3 steps:

Step 1 :

Let $M_i(t)$ be the true, unobserved longitudinal measurement at time t . Specify a cox proportional hazards model for the time-to-event outcome as:

$$\lambda(t | Z, M_i(t)) = \lambda_0(t) \exp \{ \beta^T Z + \alpha M_i(t) \}, \text{ where}$$

- $M_i(t) = \{M_i(s), 0 \leq s \leq t\}$ denotes the longitudinal history up to time t .
- α quantifies the strength of the association between the measurement and the risk of the event
- Z denotes baseline covariates.

Step 2 :

Let $Y_i(t)$ be the observed longitudinal measurements, specify a LMM as:

$$\begin{aligned} Y_i(t) &= M_i(t) + \epsilon_i^*(t) \\ &= \underbrace{\theta_0}_{\text{intercept term}} + \underbrace{X_i(t) \theta_1}_{\text{coefficient for } X_i(t)} + \underbrace{b_{0i}}_{\text{random intercept}} + \underbrace{b_{1i} X_i(t)}_{\text{random slope}} + \epsilon_i^* \end{aligned}$$

Note:

- The LMM is for $M_i(t)$, the true measurements.
- $\epsilon_i^*(t)$ denotes measurement error
- $X_i(t)$ denotes covariates, some which may be the baseline covariate Z .
- b_{0i} = random intercept
- b_{1i} = random slope.

Step 3:

Specify a joint model relating the distributions of these two processes. They are of the form:

density function

$$p(Y_i(t), T_i | \delta_i) =$$

$$\underbrace{\int p(Y_i(t) | b_i)}_{\text{pdf of } Y_i(t) \text{ conditioned on random effects}} \underbrace{\lambda(T_i | b_i)^{\delta_i} S(T_i | b_i)}_{\text{survival contribution}} \underbrace{\left\} p(b_i) db_i \right\}}_{\text{pdf of actual random effects}}, \text{ where}$$

- $b_i = \{b_{0i}, b_{1i}\}$ are the random effects that explains the interdependencies,
- $p(\cdot)$ reflects the density function.
- $S(\cdot)$ reflects the survival distribution.

// random effects explains all the interdependencies.

Note :

- For $s(T_i | b_i)$: best to assume a parametric but flexible model (e.g. splines).
- Estimation can be done using EM algorithm.
- Use R package : JM

2.5 : Missing Data

Missing data:

① Loss of efficiency

② Introduce bias

$Y_i = n \times 1$ vector of longitudinal measurements
 $= (Y_{i1}, Y_{i2}, \dots, Y_{in})^T$

$R_i = n \times 1$ vector of response indicators
 $= (R_{i1}, R_{i2}, \dots, R_{in})^T$

	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>5</u>
1	50	51	55	40	42
2	42	56	51	.	.
3	.	.	40	92	70
4	39	.	54	.	49

$$1 \quad Y_1 = [50, 51, 55, 40, 42]^T$$

$$R_1 = [1, 1, 1, 1, 1]^T$$

$$2 \quad Y_2 = [42, 56, 51, NA, NA]^T$$

$$R_2 = [1, 1, 1, 0, 0]^T$$

$$3 \quad Y_3 = [NA, NA, 40, 92, 70]^T$$

$$R_3 = [0, 0, 1, 1, 1]^T$$

$$4 \quad Y_4 = [39, NA, 54, NA, 49]^T$$

$$R_4 = [1, 0, 1, 0, 1]^T$$

Missing data mechanisms:

- ① Missing completely at random (MCAR)
- ② Missing at random (MAR)
- ③ Missing not at Random (MNAR)

Let y_i^o denote y_{ij} measurements that I observed.

y_i^m denote y_{ij} measurements that are missing.

For MCAR:

$p(\text{responses are missing})$ is unrelated to both y_i^o and y_i^m :

$$p(R_i | y_i^o, y_i^m) = p(R_i)$$

$\Rightarrow$ all methods are valid.

For MAR:

$p(\text{responses are missing})$ is related to y_i^o but unrelated to y_i^m :

$$p(R_i | y_i^o, y_i^m) = p(R_i | y_i^o)$$

$\Rightarrow$ Not completely random, not all methods are valid.

1) Mixed models with correctly specified correlation structure are valid.

2) GEEs are NOT valid.

For MNAR:

$p(\text{responses are missing})$ is related to y_i^m .

$$\begin{aligned} p(R_i | y_i^o, y_i^m) &= p(R_i | y_i^m) \\ &= p(R_i | y_i^m, y_i^o) \end{aligned}$$

$\Rightarrow$ observed data cannot be viewed as representative of population.

All statistical Methods are not valid.

$\Rightarrow$ joint distribution of $\{R_i, y_i^o, y_i^m\}$

Handling Missing Data :

- ① Complete-case analysis : ignores subject with incomplete data.
- ② Available data : uses all available data.
 - ⇒ only valid under MCAR.
 - ⇒ sometimes valid for MAR.

③ Imputation. Many ways.

3.1) Last observation carry forward.

variations : carry forward the

· Baseline.

· worst observation.

3.2) Multiple imputation. Most appropriate.

→ Missing values are replaced with a set of m plausible values creating m filled-in versions of the dataset,

→ the analysis is done with each of the m datasets,

→ the estimates for each of these m analyses are combined together.

Let $\hat{\beta}^{(k)}$ denote the estimation of β from the k^{th} filled-in dataset (for $k = 1, 2, \dots, m$). The single estimate of β is given by:

$$\bar{\beta} = \frac{1}{m} \sum_{k=1}^m \hat{\beta}^{(k)}$$

Estimated variance of $\bar{\beta}$:

$$\frac{1}{m} \sum_{k=1}^m \hat{\text{Cov}}(\hat{\beta}^{(k)}) + \left(1 + \frac{1}{m}\right) \left(\frac{1}{m-1}\right) \sum_{k=1}^m (\hat{\beta}^{(k)} - \bar{\beta})(\hat{\beta}^{(k)} - \bar{\beta})^T$$

There are many options for imputation; generally the idea behind them is to draw values of y_i^m from the conditional distribution $f(y_i^m | y_i^o, x_i)$ where x_i denotes covariates.

3.2: Predictive mean matching:

series of regression models to model the missing y

1	2	3	4	5
$\frac{1}{x}$	$\frac{1}{x}$	$\frac{1}{x}$	$\frac{1}{x}$	$\frac{1}{0}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	x
$\vdots$	$\vdots$	$\vdots$	$\vdots$	x
$\vdots$	$\vdots$	$\vdots$	$\vdots$	0
				x
				0
				x

$$E(y_{i5}) = a_0 + a_1 y_{i1} + a_2 y_{i2} + a_3 y_{i3} + a_4 y_{i4} + a_5 x_i$$